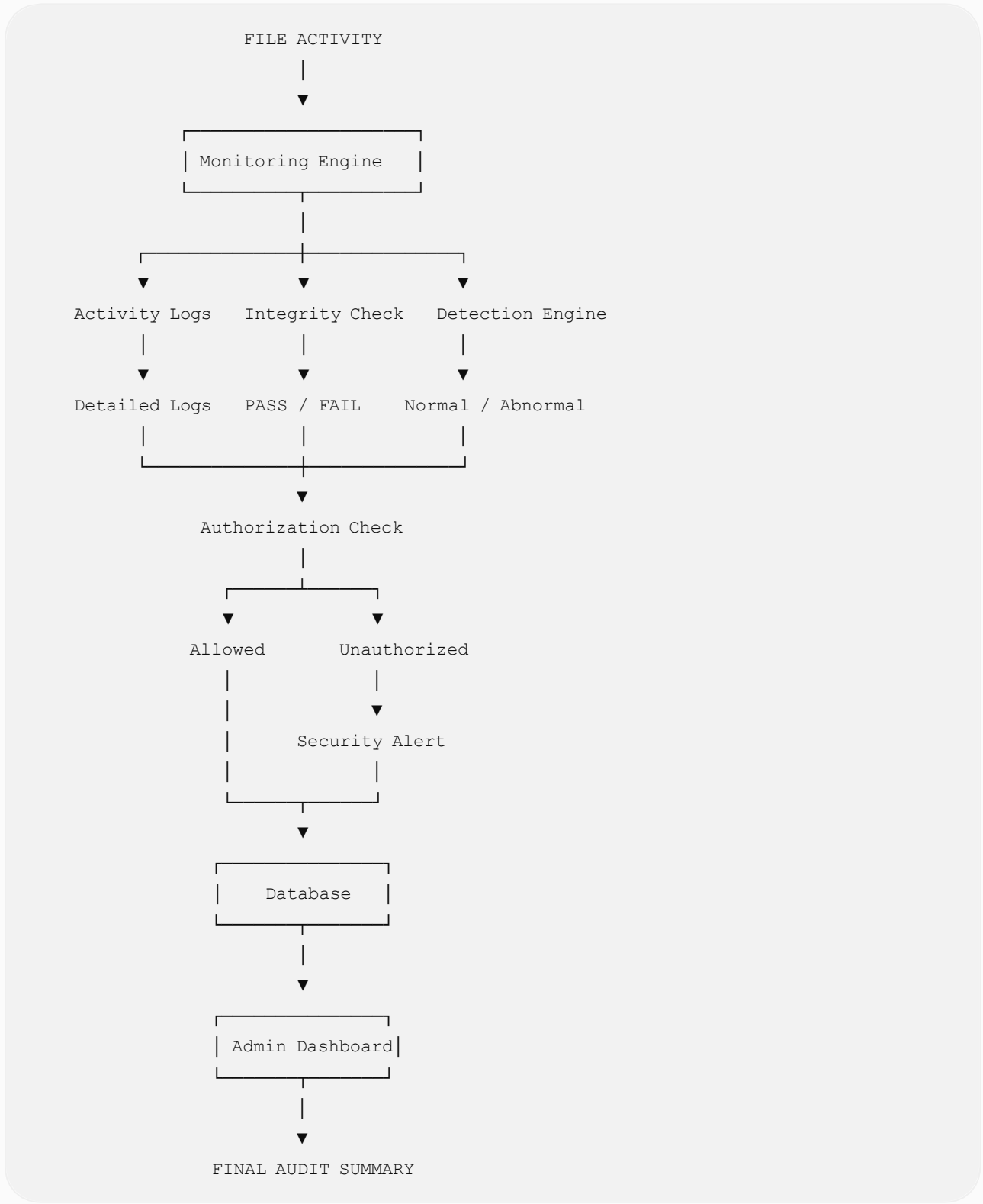


**Secure File Transfer Monitoring System**, the output should be designed around **five major security outputs**. This makes the project clear and measurable for an internship report.

## System Output Architecture



### 1. Detailed File Activity Logs

The first major output is a complete record of file activity.

#### Example

| Field          | Output               |
|----------------|----------------------|
| Event ID       | EVT-00125            |
| Date/Time      | 15-09-2026 14:05:22  |
| User           | employee01           |
| File           | confidential.xlsx    |
| Operation      | COPY                 |
| Source         | C:\Company\Documents |
| Destination    | E:\USB               |
| File Size      | 2.5 MB               |
| Classification | Sensitive            |
| Authorization  | Denied               |
| Risk           | High                 |
| Status         | Alert Generated      |

The system should record activities such as:

- Create
- Copy
- Move
- Modify
- Delete
- Upload
- Download

This provides an **audit trail** of file movement.

---

## 2. Unauthorized Movement Alerts

When a user moves or copies a file without permission, the system generates an alert.

### Example output

| UNAUTHORIZED FILE MOVEMENT |                     |
|----------------------------|---------------------|
| User                       | : employee01        |
| File                       | : confidential.xlsx |
| Source                     | : Company/Documents |
| Destination                | : External USB      |
| Authorization              | : DENIED            |
| Severity                   | : HIGH              |
| Time                       | : 14:05:22          |

The dashboard can classify alerts as:

LOW  
MEDIUM  
HIGH  
CRITICAL

### 3. Integrity Check Status

The system should show whether the transferred file has maintained its integrity.

The system calculates a **SHA-256 hash** before and after the monitored transfer.

#### Successful integrity check

File: report.pdf

Original SHA-256:

A8F21C...92BD

Transferred SHA-256:

A8F21C...92BD

Integrity Status: ✓ PASS

#### Failed integrity check

File: confidential.pdf

Original SHA-256:

A8F21C...92BD

Transferred SHA-256:

7C92AB...41EF

Integrity Status: X FAIL  
Risk: HIGH

The dashboard can summarize:

| Integrity Status | Count |
|------------------|-------|
| PASS             | 1,180 |
| FAIL             | 12    |
| Not Checked      | 58    |

A failed hash comparison should create a security event for investigation.

## 4. Detection of Abnormal File Transfers

The system should identify file transfers that differ from normal or approved behavior.

### Examples of abnormal activity

- Large number of files transferred in a short period
- Sensitive file transferred to an unauthorized location
- Executable files transferred unexpectedly
- Transfer outside approved hours
- User accessing unusual directories
- Repeated authorization failures
- File hash mismatch
- Large-volume transfer to removable media

### Example

```
Normal Activity

employee01
  ↓
5 files
  ↓
Approved server
  ↓
ALLOWED
```

Abnormal activity:

```
employee01
  ↓
250 files
  ↓
External USB
  ↓
Sensitive files
  ↓
Outside normal working hours
  ↓
ABNORMAL TRANSFER DETECTED
```

## Detection output

```
|| ABNORMAL TRANSFER DETECTED ||
||                               ||
|| User      : employee01      ||
|| Files     : 250              ||
|| Destination: External USB    ||
|| Pattern   : Excessive transfer volume ||
|| Risk      : HIGH             ||
|| Status    : INVESTIGATION REQUIRED ||
||                               ||
```

## 5. Final Audit Summary

The final output should provide management/security personnel with a summary of all monitored activity.

### Example Final Audit Report

```
=====
SECURE FILE TRANSFER AUDIT SUMMARY
=====

Reporting Period:
01-09-2026 to 15-09-2026

Total File Events      : 1,250
Authorized Transfers   : 1,100
Unauthorized Transfers : 150

Sensitive File Transfers : 85
Abnormal Transfers      : 42
```

Integrity Checks

|             |         |
|-------------|---------|
| -----       |         |
| Passed      | : 1,180 |
| Failed      | : 12    |
| Not Checked | : 58    |

Security Alerts

|          |      |
|----------|------|
| -----    |      |
| Low      | : 15 |
| Medium   | : 18 |
| High     | : 7  |
| Critical | : 2  |

Top Security Incidents

- 
- Unauthorized USB transfer
  - Excessive file movement
  - Sensitive file transfer
  - SHA-256 integrity mismatch

Overall Security Status:  
HIGH RISK - REVIEW REQUIRED

=====

Recommended Dashboard

Your Flask web application can have these sections:

|                                        |      |              |        |                 |
|----------------------------------------|------|--------------|--------|-----------------|
| SECURE FILE TRANSFER MONITORING SYSTEM |      |              |        |                 |
|                                        |      |              |        |                 |
| Total Events                           |      | Unauthorized | Alerts |                 |
| 1250                                   |      | 150          | 42     |                 |
|                                        |      |              |        |                 |
| FILE ACTIVITY LOGS                     |      |              |        |                 |
| User                                   | File | Operation    | Source | Destination     |
|                                        |      |              |        |                 |
| UNAUTHORIZED MOVEMENT ALERTS           |      |              |        |                 |
| High-risk and critical activities      |      |              |        |                 |
|                                        |      |              |        |                 |
| INTEGRITY STATUS                       |      |              |        |                 |
| ✓ PASS: 1180                           |      | ✗ FAIL: 12   |        |                 |
|                                        |      |              |        |                 |
| ABNORMAL TRANSFER DETECTION            |      |              |        |                 |
| Suspicious patterns and risk scores    |      |              |        |                 |
|                                        |      |              |        |                 |
| FINAL AUDIT SUMMARY                    |      |              |        |                 |
| Statistics                             |      | Incidents    | Risk   | Recommendations |

## Final Project Output

Your project can therefore claim these **five core outputs**:

1. **Detailed File Activity Logs** — complete record of file operations and transfers.
2. **Unauthorized Movement Alerts** — identifies and reports prohibited file movements.
3. **Integrity Check Status** — displays SHA-256 verification as **PASS/FAIL**.
4. **Abnormal File Transfer Detection** — identifies suspicious transfer patterns and assigns risk.
5. **Final Audit Summary** — consolidates events, alerts, integrity results, incidents, and security recommendations into a report.

This gives your project a clear **input** → **monitoring** → **analysis** → **detection** → **alerting** → **reporting** lifecycle suitable for a cybersecurity internship project.